Computer Vision:

Week-3 (02-06 Feb) Theory Lectures

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Review

- KeyPoints- Edges –Corners
- Harris & FAST edge detectors
- Characteristics of mentioned Edge Detectors
- $| I(x_i) - I(p) | > t$
- Brighter & Darker Conditions

In Computer Vision

- Match images
- Recognize objects
- Track motion
- Do image stitching (panorama)
- Perform SLAM & AR

A feature is a distinctive and repeatable point or region in an image.

Good features should be:

- Invariant to scale
- Invariant to rotation
- Robust to illumination change
- Distinctive and repeatable

Invariant to scale/rotation

- Invariant → does NOT change, even if something else changes.

-In computer vision:

- The image may change
- But the feature we detect should remain the same

Invariant to scale

- If an object becomes bigger or smaller,
- we should still recognize it.

A car near you (big)
The same car far away (small)

Invariant to scale

- If a feature detector is scale invariant:
- It detects the same keypoints
- Even if the image is zoomed in or zoomed out.

Rotation Invariance (Invariant to Rotation)

- If an object is rotated, we should still recognize it.

-***The letter A:***

- **Normal A**
- **Rotated A (tilted)**

If a feature detector/descriptor is rotation invariant:

- *It finds the same feature*
- *Even if the image is rotated*

SIFT – Scale Invariant Feature Transform

Scale-Space Construction

- Detect stable keypoints across different scales and rotations

SIFT uses Gaussian Blur:

$$L(x, y, \sigma) = G(x, y, \sigma) * I(x, y)$$

Creates multiple blurred images at different σ (scales)

Objects appear different at different sizes → scale invariance

SIFT – Scale Invariant Feature Transform

Difference of Gaussian (DoG)

- Approximation of Laplacian of Gaussian (LoG)

Detect blob-like structures

$$DoG(x, y, \sigma) = L(x, y, k\sigma) - L(x, y, \sigma)$$

$$The\ image\ blurred\ at\ a\ higher\ scale\ \left(more\ blurry\right),$$
$$where\ k\ is\ a\ constant\ multiplier.$$

Keypoint Localization

$\textit{Find } \textbf{local maxima / minima} \textit{ in DoG}$

$\textit{Compare pixel with} \textbf{26 neighbors}$

$\textit{Remove: Low contrast points}$

$\textit{Edge responses}$

Orientation Assignment

Assign dominant orientation using gradient magnitude & direction

$$Magnitude(m) = \sqrt{L_x^2 + L_y^2}$$

$$Orientation(\theta) = \tan^{-1}(L_y / L_x)$$

Feature Descriptor

$$16\times16\,Neighborhood: Around\ a\ detected\ keypoint,$$
$$the\ algorithm\ looks\ at\ a\ square\ grid\ of\ 16\times16\ pixels.$$

Feature Descriptor

Divided into 4×4 Blocks:

This large grid is subdivided into 16 smaller windows (each containing $4 \times 4\,pixels$)

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Each block → **8-bin orientation histogram**

$$16 \text{ blocks} \times 8 \text{ directions per block} = 128 \text{ total values}$$

Feature Detection vs Feature Description

Step	*Purpose*
Detection	Find interesting keypoints
Description	Represent keypoints numerically
Matching	Compare descriptors between images

SURF – Speeded Up Robust Features

-Purpose

- *SURF is used to find and match important points in images quickly.*

SIFT was accurate but slow → SURF makes it faster

SURF finds special points (blobs) in an image that stay the same even if the image is rotated or resized

How SURF Finds Keypoints (Very Easy)

✔ SURF looks for blob-shaped regions

✔ It uses the Hessian Matrix (just a math tool to find blobs)

High Hessian value = Strong feature point

SURF uses:

- *Integral Images → very fast calculations*
- *Box filters instead of heavy Gaussian filters*

- *This makes SURF much faster than SIFT*

Finds direction of the keypoint

- *Uses Haar wavelets (fast intensity changes)*